Week 1 Day 1 Summary Report

Problem Framing:

This is a binary classification problem where the goal is to predict whether an individual earns more than $50K per year. The positive class is income >50K, while the negative class is income <=50K. The business objective is to identify individuals likely to earn more than $50K so marketing efforts can focus on higher-income customers and reduce wasted outreach. The class distribution is imbalanced: the positive class represents 23.93% of the data, while the negative class represents 76.07%.

Chosen Metric:

F1 score is the primary evaluation metric because it balances precision and recall. Precision matters because false positives lead to wasted outreach, while recall matters because we do not want to miss too many potential high-income customers. We also monitor precision and recall to understand the trade-off between wasted outreach and missed opportunities.

Baseline Results:

The majority-class baseline predicts every observation as <=50K and achieves 76.07% accuracy, but this is misleading because it produces 0.00 precision, 0.00 recall, and 0.00 F1 for the positive class. This illustrates the risk of using accuracy on an imbalanced dataset. The single-feature education baseline uses education-num >= 13 and performs better: Accuracy = 0.768, Precision = 0.548, Recall = 0.411, F1 = 0.491, ROC AUC = 0.719, and PR AUC = 0.444. This shows that education level contains meaningful signal for income prediction, but the rule is still far from sufficient.

Error Analysis:

The sampled false positives and false negatives reveal the limitations of the simple education rule. False positives are often individuals with strong educational backgrounds or professional-looking profiles who still earn <=50K, suggesting that education alone is not enough to distinguish high earners from lower earners. False negatives are often younger individuals, people working very long hours, or individuals who do not fit the standard educational path but still earn above $50K. This indicates that the model is capturing a narrow pattern and missing more nuanced cases driven by work intensity, role type, and other interacting factors.

Next Steps:

To improve performance, we should address several data and feature issues. These include handling missing and inconsistent categorical values in workclass, occupation, and native-country; encoding nominal and high-cardinality categorical variables more effectively; applying log transformations to highly skewed numerical features like capital-gain and capital-loss; scaling continuous features such as age, fnlwgt, and hours-per-week; engineering interaction features such as age × hours-per-week and net capital; and grouping low-frequency categories to reduce sparsity. These changes will help the model learn more robust and generalizable patterns.

Conclusion:

The primary metric to optimize for the rest of the week is F1 score, not accuracy. The reason is that the dataset is imbalanced and the majority baseline appears strong on accuracy while failing completely on the positive class. The education rule improves on this baseline with an F1 score of 0.491, but there is still significant room for improvement. Since the business goal is to identify high-income individuals while minimizing wasted outreach, F1 is the correct target because it balances precision and recall and directly reflects whether we are succeeding at the actual prediction task.